Scott's Electrical Variation Response Equation

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Abstract

This paper introduces and investigates a parameterized mathematical functional designed to represent an accumulated electrical response to historical voltage variations.

The proposed Electrical Variation Response Equation (EVRE) is defined by

$$\Omega_s(t) = \sqrt{\Omega_0^2 + \frac{\beta}{T}\int_0^t e^{-\lambda(t-\tau)}\left(\frac{dV(\tau)}{d\tau}\right)^2 d\tau},$$

where Ω_0 represents an initial response, $V(t)$ denotes a time-dependent voltage signal, β is a positive scaling parameter, T is a reference time interval, and λ is a positive memory-decay parameter.

The proposed functional combines squared voltage variation, exponential temporal weighting, and a nonlinear square-root transformation.

We investigate its dimensional consistency, nonnegativity, boundedness under specified input conditions, continuity, and dependence on its parameters.

An equivalent first-order differential equation is derived, allowing the historical integral to be represented through a local evolution equation. We also examine the behavior under constant voltage, scaled voltage inputs, and bounded voltage variation.

A reconstruction expression is obtained for the magnitude of the instantaneous voltage derivative when the response and model parameters are known.

The proposed equation is a mathematical construction rather than an established electrical law. Its physical interpretation, practical usefulness, and empirical significance require further investigation.

Keywords: Electrical response; voltage variation; exponential memory; integral functionals;

nonlinear mathematical modeling; electrical systems; parameter sensitivity; differential equations.

1 Introduction

Electrical systems are commonly investigated through relationships involving voltage, current, resistance, capacitance, and inductance.

Many electrical quantities are described using instantaneous relationships between circuit variables. However, mathematical models can also be constructed to investigate quantities that depend on the previous behavior of a signal.

A time-dependent voltage signal may experience continuous variation during the operation of an electrical system.

The magnitude and frequency of these variations can be investigated mathematically through derivatives and integral expressions.

This paper introduces a parameterized functional that accumulates squared voltage variation over a historical time interval while assigning exponentially decreasing weight to older changes.

The resulting expression introduces a memory-dependent mathematical response quantity denoted by $\Omega_s(t)$.

The purpose of this investigation is to establish the mathematical properties of the proposed functional and determine whether it provides a useful framework for describing accumulated electrical variation.

The principal questions considered are:

1. Is the proposed functional mathematically well-defined and dimensionally consistent?
2. Is its response always nonnegative?
3. Under what conditions is the response bounded?
4. How does the memory parameter influence the contribution of historical voltage changes?
5. Can the integral representation be converted into a differential equation?
6. Is information about the instantaneous voltage variation recoverable from the response?

The investigation begins with the definition of the mathematical quantities and proceeds toward the construction of the proposed equation.

2 Mathematical Foundations

2.1 The voltage function

Let

$$V : [0, \infty) \to \mathbb{R}$$

represent a time-dependent voltage signal.

The voltage is assumed to be locally absolutely continuous, ensuring that its derivative exists almost everywhere and that the corresponding integral expressions can be investigated.

Define the instantaneous voltage variation by

$$v'(t) = \frac{dV(t)}{dt}. \tag{1}$$

The derivative measures the rate at which the voltage changes with respect to time.

Its units are

$$[v'(t)] = \frac{\mathrm{V}}{\mathrm{s}}.$$

The square of this quantity is used in the construction of the proposed functional.

2.2 The historical weighting function

Introduce an exponential weighting function:

$$K(t, \tau) = e^{-\lambda(t-\tau)}, \tag{2}$$

where

$$\lambda > 0$$

is the memory-decay parameter.

For

$$0 \leq \tau \leq t,$$

the weighting function satisfies

$$0 < K(t, \tau) \leq 1.$$

At the present time, where $\tau = t$,

$$K(t, t) = 1.$$

For earlier times, the weighting function decreases as the difference $t - \tau$ increases.

Thus, recent voltage variations receive greater weight than older voltage variations.

2.3 The accumulated variation functional

Define the accumulated historical variation by

$$Q(t) = \int_0^t e^{-\lambda(t-\tau)} \left(\frac{dV(\tau)}{d\tau} \right)^2 d\tau. \tag{3}$$

This quantity represents an exponentially weighted accumulation of squared voltage variation.

Since the integrand is nonnegative,

$$Q(t) \geq 0.$$

The functional $Q(t)$ provides the mathematical foundation for the proposed response equation.

3 Definition of the Electrical Variation Response Equation

3.1 Construction

The accumulated variation functional provides a nonnegative quantity that depends on the historical behavior of a voltage signal.

To construct a response quantity with units of electrical resistance, introduce the following parameters:

- Ω_0: Initial response value.
- β: Positive scaling coefficient.
- T: Positive reference time interval.
- λ: Positive memory-decay parameter.

The parameters satisfy

$$\Omega_0 \geq 0, \qquad \beta > 0, \qquad T > 0, \qquad \lambda > 0.$$

3.2 Formal definition

Definition 3.1 (Electrical Variation Response Equation)**.** For an admissible voltage signal $V(t)$, define the Electrical Variation Response Equation by

$$\Omega_s(t) = \sqrt{\Omega_0^2 + \frac{\beta}{T} \int_0^t e^{-\lambda(t-\tau)} \left(\frac{dV(\tau)}{d\tau} \right)^2 d\tau}. \tag{4}$$

The quantity $\Omega_s(t)$ is defined as a memory-dependent electrical response measure.

It is not automatically equivalent to the physical resistance of a component.

Its interpretation as a resistance-related quantity would require additional physical justification.

3.3 Dimensional consistency

The proposed expression must be dimensionally consistent.

The voltage derivative has dimensions

$$\left[\frac{dV}{dt}\right] = \frac{\mathrm{V}}{\mathrm{s}}.$$

Therefore,

$$\left[\left(\frac{dV}{dt}\right)^2 dt\right] = \frac{\mathrm{V}^2}{\mathrm{s}}.$$

The exponential function is dimensionless provided that

$$[\lambda] = \mathrm{s}^{-1}.$$

Since the quantity inside the square root must have units of resistance squared, the scaling parameter must satisfy

$$[\beta] = \frac{\Omega^2 \mathrm{s}^2}{\mathrm{V}^2}. \tag{5}$$

The reference time has units

$$[T] = \mathrm{s}.$$

Consequently,

$$\left[\frac{\beta}{T} Q(t)\right] = \Omega^2.$$

Both terms inside the square root therefore have identical dimensions.

The proposed equation is dimensionally consistent under these parameter definitions.

3.4 Equivalent representation

Using Equation (3), the response equation can be written as

$$\Omega_s(t) = \sqrt{\Omega_0^2 + \frac{\beta}{T} Q(t)}. \tag{6}$$

This representation will be useful when investigating the mathematical properties of the functional.

4 Basic Mathematical Properties

4.1 Nonnegativity

Theorem 4.1. *For every admissible voltage signal and all permitted parameter values,*

$$\Omega_s(t) \geq \Omega_0 \geq 0. \tag{7}$$

Proof. From Equation (3),

$$Q(t) = \int_0^t e^{-\lambda(t-\tau)} \left(V'(\tau)\right)^2 d\tau.$$

The exponential weighting function is positive, and the squared voltage derivative is nonnegative.

Therefore,

$$Q(t) \geq 0.$$

Since

$$\beta > 0, \qquad T > 0,$$

we obtain

$$\frac{\beta}{T}Q(t) \geq 0.$$

It follows that

$$\Omega_0^2 + \frac{\beta}{T}Q(t) \geq \Omega_0^2.$$

Taking nonnegative square roots proves

$$\Omega_s(t) \geq \Omega_0.$$

□

4.2 Initial condition

Proposition 4.2. *The response equation satisfies*

$$\Omega_s(0) = \Omega_0. \tag{8}$$

Proof. At $t = 0$, the integral has identical upper and lower limits.

Therefore,

$$Q(0) = 0.$$

Substituting into Equation (6) gives

$$\Omega_s(0) = \sqrt{\Omega_0^2}.$$

Since $\Omega_0 \geq 0$,

$$\Omega_s(0) = \Omega_0.$$

□

4.3 Equality condition

Proposition 4.3. *For a fixed time $t > 0$,*

$$\Omega_s(t) = \Omega_0$$

if and only if

$$V'(\tau) = 0$$

for almost every $\tau \in [0, t]$.

Proof. The equality

$$\Omega_s(t) = \Omega_0$$

is equivalent to

$$Q(t) = 0.$$

Because the weighting function is strictly positive and the squared derivative is nonnegative, the integral can equal zero only when

$$V'(\tau) = 0$$

almost everywhere on the integration interval.

The converse follows immediately by substitution.

□

4.4 Continuity

Theorem 4.4. *If V' is locally square-integrable, then $\Omega_s(t)$ is continuous on every finite time interval.*

Proof. Let

$$f(\tau) = \left(V'(\tau)\right)^2.$$

By assumption, f is locally integrable.

Rewrite the accumulated functional as

$$Q(t) = e^{-\lambda t} \int_0^t e^{\lambda \tau} f(\tau) d\tau.$$

The integrand is locally integrable.

Its indefinite integral is therefore absolutely continuous on every finite interval.

Multiplication by the continuous function $e^{-\lambda t}$ preserves continuity.

Thus, $Q(t)$ is continuous.

Finally,

$$\Omega_s(t) = \sqrt{\Omega_0^2 + \frac{\beta}{T} Q(t)}$$

is a continuous composition of continuous functions.

Therefore, the response is continuous.

□

5 Differential Representation

5.1 Derivation of the differential equation

The integral representation contains information about the entire voltage history.

However, it can also be expressed through a first-order differential equation.

Recall that

$$Q(t) = e^{-\lambda t} \int_0^t e^{\lambda \tau} \left(V'(\tau)\right)^2 d\tau.$$

Differentiate using the product rule and the fundamental theorem of calculus.

We obtain

$$Q'(t) = -\lambda e^{-\lambda t} \int_0^t e^{\lambda \tau} \left(V'(\tau)\right)^2 d\tau + \left(V'(t)\right)^2 .$$

Recognizing the original integral gives

$$Q'(t) = \left(V'(t)\right)^2 - \lambda Q(t). \tag{9}$$

The initial condition is

$$Q(0) = 0.$$

This differential equation is equivalent to the original historical integral under the stated assumptions.

5.2 Response evolution

From

$$\Omega_s^2(t) = \Omega_0^2 + \frac{\beta}{T} Q(t),$$

differentiate both sides:

$$2\Omega_s(t)\Omega_s'(t) = \frac{\beta}{T} Q'(t).$$

Substituting Equation (9) gives

$$2\Omega_s \Omega_s' = \frac{\beta}{T} \left[(V')^2 - \lambda Q\right].$$

Since

$$Q = \frac{T}{\beta} \left(\Omega_s^2 - \Omega_0^2\right),$$

we obtain

$$\Omega_s'(t) = \frac{\beta(V'(t))^2 - \lambda T\left(\Omega_s^2(t) - \Omega_0^2\right)}{2T\Omega_s(t)}. \tag{10}$$

This equation holds wherever the response is differentiable and strictly positive.

The differential representation shows that the response is influenced by two competing contributions.

The first is the instantaneous voltage variation.

The second is the decay of accumulated historical variation.

5.3 Interpretation of the differential representation

The term

$$\beta(V'(t))^2$$

represents the instantaneous contribution from voltage variation.

The term

$$\lambda T\left(\Omega_s^2(t) - \Omega_0^2\right)$$

represents the decay of the accumulated response contribution.

Consequently, the response does not necessarily increase at every moment.

It can decrease when historical variation decays faster than new variation is introduced.

6 Boundedness and Input Conditions

6.1 Bounded voltage derivative

Consider an admissible voltage signal satisfying

$$|V'(t)| \leq M \tag{11}$$

for some finite constant $M \geq 0$.

Then

$$(V'(t))^2 \leq M^2.$$

Substitution into the accumulated functional gives

$$Q(t) \leq M^2 \int_0^t e^{-\lambda(t-\tau)} d\tau.$$

Evaluating the integral,

$$Q(t) \leq \frac{M^2}{\lambda} \left(1 - e^{-\lambda t}\right).$$

Therefore, the response satisfies the following result.

Theorem 6.1. *If the voltage derivative is bounded by M, then*

$$\Omega_0 \leq \Omega_s(t) \leq \sqrt{\Omega_0^2 + \frac{\beta M^2}{T\lambda} (1 - e^{-\lambda t})}. \tag{12}$$

Proof. The previous inequality establishes that

$$0 \leq Q(t) \leq \frac{M^2}{\lambda} \left(1 - e^{-\lambda t}\right).$$

Multiplying by β/T and adding Ω_0^2 gives

$$\Omega_0^2 \leq \Omega_s^2(t) \leq \Omega_0^2 + \frac{\beta M^2}{T\lambda} \left(1 - e^{-\lambda t}\right).$$

Taking square roots proves the result.

□

6.2 Long-time boundedness

Taking the limit as $t \to \infty$ in Equation (12) gives

$$\limsup_{t\to\infty} \Omega_s(t) \leq \sqrt{\Omega_0^2 + \frac{\beta M^2}{T\lambda}}. \tag{13}$$

Thus, bounded voltage variation produces a uniformly bounded response.

6.3 Constant voltage

Suppose that the voltage remains constant:

$$V(t) = V_0.$$

Then

$$V'(t) = 0.$$

Consequently,

$$Q(t) = 0$$

and

$$\Omega_s(t) = \Omega_0. \tag{14}$$

A constant voltage therefore produces no additional accumulated response.

7 Parameter Sensitivity

The response depends on three principal model parameters: β, T, and λ.

Their effects are examined independently.

7.1 Sensitivity with respect to β

Differentiating Equation (6) with respect to β gives

$$\frac{\partial \Omega_s}{\partial \beta} = \frac{Q(t)}{2T\Omega_s(t)}. \tag{15}$$

For positive accumulated variation,

$$Q(t) > 0.$$

Therefore,

$$\frac{\partial \Omega_s}{\partial \beta} > 0.$$

Increasing β increases the modeled response whenever voltage variation has occurred.

7.2 Sensitivity with respect to T

Differentiating with respect to T gives

$$\frac{\partial \Omega_s}{\partial T} = -\frac{\beta Q(t)}{2T^2\Omega_s(t)}. \tag{16}$$

For positive accumulated variation,

$$\frac{\partial \Omega_s}{\partial T} < 0.$$

Thus, increasing the reference interval decreases the modeled response when the other parameters and input history remain fixed.

7.3 Sensitivity with respect to λ

Differentiate the historical functional:

$$Q(t) = \int_0^t e^{-\lambda(t-\tau)}(V'(\tau))^2 d\tau.$$

Using differentiation under the integral sign,

$$\frac{\partial Q}{\partial \lambda} = -\int_0^t (t-\tau)e^{-\lambda(t-\tau)}(V'(\tau))^2 d\tau.$$

Therefore,

$$\frac{\partial \Omega_s}{\partial \lambda} = -\frac{\beta}{2T\Omega_s}\int_0^t (t-\tau)e^{-\lambda(t-\tau)}(V'(\tau))^2 d\tau. \tag{17}$$

The integrand is nonnegative.

Consequently,

$$\frac{\partial \Omega_s}{\partial \lambda} \leq 0.$$

Increasing the memory-decay parameter decreases the contribution of historical voltage variation.

7.4 Parameter interpretation

The three parameters influence different aspects of the proposed model.

- β controls the magnitude of the accumulated contribution.
- T establishes a reference time scale for the response.
- λ controls the rate at which historical contributions lose weight.

These parameters provide independent mechanisms for adjusting the mathematical behavior of the functional.

8 Response to Particular Voltage Signals

8.1 Constant voltage slope

Consider a voltage signal of the form

$$V(t) = V_0 + at,$$

where a is a constant voltage slope.

Then

$$V'(t) = a.$$

Substituting into Equation (3),

$$Q(t) = a^2 \int_0^t e^{-\lambda(t-\tau)} d\tau.$$

Therefore,

$$Q(t) = \frac{a^2}{\lambda} \left(1 - e^{-\lambda t}\right).$$

The response becomes

$$\Omega_s(t) = \sqrt{\Omega_0^2 + \frac{\beta a^2}{T\lambda} (1 - e^{-\lambda t})}. \tag{18}$$

As time increases, the response approaches

$$\lim_{t \to \infty} \Omega_s(t) = \sqrt{\Omega_0^2 + \frac{\beta a^2}{T\lambda}}. \tag{19}$$

8.2 Sinusoidal voltage

Consider a sinusoidal voltage signal:

$$V(t) = A\sin(\omega t).$$

Its derivative is

$$V'(t) = A\omega\cos(\omega t).$$

Therefore,

$$(V'(t))^2 = A^2\omega^2\cos^2(\omega t).$$

Using the identity

$$\cos^2(\omega t) = \frac{1+\cos(2\omega t)}{2},$$

the accumulated functional becomes

$$Q(t) = \frac{A^2\omega^2}{2}\int_0^t e^{-\lambda(t-\tau)}\left[1+\cos(2\omega\tau)\right]d\tau.$$

Evaluating the integral gives

$$Q(t) = \frac{A^2\omega^2}{2}\left[\frac{1-e^{-\lambda t}}{\lambda}\right. \tag{20}$$

$$\left. + \frac{\lambda\cos(2\omega t) + 2\omega\sin(2\omega t) - \lambda e^{-\lambda t}}{\lambda^2+4\omega^2}\right]. \tag{21}$$

Substitution into the original response equation provides the corresponding sinusoidal response.

This demonstrates that the proposed functional can be evaluated for different classes of time-dependent voltage signals.

9 Scaling Properties

9.1 Voltage amplitude scaling

Consider a scaled voltage signal

$$\widetilde{V}(t) = aV(t),$$

where a is a real constant.

Then

$$\widetilde{V}'(t) = aV'(t).$$

The accumulated functional becomes

$$\widetilde{Q}(t) = a^2 Q(t).$$

Consequently,

$$\widetilde{\Omega}_s(t) = \sqrt{\Omega_0^2 + a^2\left(\Omega_s^2(t) - \Omega_0^2\right)}. \tag{22}$$

Thus, the accumulated contribution scales quadratically with voltage amplitude before the square-root transformation.

9.2 Voltage offset invariance

Consider adding a constant voltage offset:

$$\widetilde{V}(t) = V(t) + C.$$

Differentiation gives

$$\widetilde{V}'(t) = V'(t).$$

Therefore,

$$\widetilde{\Omega}_s(t) = \Omega_s(t). \tag{23}$$

The proposed functional is invariant under constant voltage offsets.

Only changes in voltage contribute to the accumulated response.

10 Reconstruction of Voltage Variation

10.1 Derivation

The differential representation provides a method for recovering the magnitude of the instantaneous voltage derivative.

Starting from Equation (10),

$$2T\Omega_s\Omega_s' = \beta(V')^2 - \lambda T(\Omega_s^2 - \Omega_0^2).$$

Rearranging,

$$\beta(V')^2 = 2T\Omega_s\Omega_s' + \lambda T(\Omega_s^2 - \Omega_0^2).$$

Therefore,

$$|V'(t)| = \sqrt{\frac{2T\Omega_s(t)\Omega_s'(t) + \lambda T\left(\Omega_s^2(t) - \Omega_0^2\right)}{\beta}}. \tag{24}$$

This expression is valid wherever the required derivatives exist and the expression under the square root is nonnegative.

10.2 Limitations of reconstruction

The reconstruction expression recovers the magnitude of the voltage derivative.

It does not independently recover its sign.

Consequently, the response alone cannot distinguish between equal-magnitude positive and negative voltage variations.

This is an important limitation of the proposed functional.

11 Mathematical Interpretation and Potential Applications

11.1 Mathematical interpretation

The Electrical Variation Response Equation combines three mathematical operations.

First, the voltage derivative measures instantaneous variation.

Second, the squared derivative produces a nonnegative contribution.

Third, the exponential kernel assigns decreasing weight to older contributions.

The resulting integral is transformed through a square root and combined with an initial response value.

The equation therefore defines a nonnegative memory-dependent functional of the voltage history.

11.2 Possible electrical applications

Potential areas for further investigation include:

1. Mathematical analysis of voltage variation in time-dependent electrical systems.
2. Investigation of voltage transient histories in circuit models.
3. Comparison of different voltage waveforms through their accumulated variation responses.
4. Mathematical investigation of historical voltage dependence.
5. Development of experimental methods for estimating the model parameters.

These applications are prospective.

The functional has not been established as a direct measurement of physical resistance, component degradation, or circuit instability.

Such interpretations would require independent physical derivations and experimental validation.

12 Limitations and Further Investigation

The mathematical properties established in this paper do not independently demonstrate that the proposed response corresponds to a new physical electrical quantity.

Several questions remain open.

12.1 Physical interpretation

The quantity $\Omega_s(t)$ has units of ohms, but dimensional consistency alone does not establish that it represents electrical resistance.

A physical interpretation would require a relationship between the functional and measurable circuit behavior.

12.2 Parameter identification

The parameters β, T, and λ must be investigated to determine whether they can be reliably estimated from experimental voltage histories.

Their practical identifiability has not been established.

12.3 Comparison with existing models

The exponential memory kernel and squared derivative are established mathematical constructions.

Therefore, the proposed combination should be compared with existing integral functionals, signal-processing methods, and electrical system models.

Such comparisons are necessary before any claim of mathematical or practical novelty can be justified.

12.4 Experimental validation

Future investigations could compare the predicted response against measurements obtained from controlled voltage signals.

Such experiments would be necessary to determine whether the functional provides information beyond existing electrical measurements.

13 Conclusion

This paper introduced the Electrical Variation Response Equation, a parameterized mathematical functional constructed from the historical variation of a voltage signal.

The proposed equation is

$$\Omega_s(t) = \sqrt{\Omega_0^2 + \frac{\beta}{T}\int_0^t e^{-\lambda(t-\tau)}\left(\frac{dV(\tau)}{d\tau}\right)^2 d\tau}.$$

The analysis established its dimensional consistency under appropriate parameter units, nonnegativity, continuity, and boundedness under bounded voltage-derivative conditions.

An equivalent first-order differential equation was derived, demonstrating that the historical integral can be represented through a local evolution equation.

The investigation also established parameter sensitivities, voltage scaling properties, constant-offset invariance, and expressions for particular voltage signals.

A reconstruction formula was obtained for the magnitude of the instantaneous voltage derivative when the response and model parameters are known.

The proposed functional provides a mathematical framework for examining accumulated voltage variation with exponentially decreasing historical weighting.

However, its physical significance and practical usefulness remain open questions.

Further work should investigate its relationship to existing electrical models, parameter identifiability, and experimental validation.

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